- No lecture on Friday 14 March

- A3 deadline

- Have received some A1 + test remark queries. I will attend to those, but after other tasks completed.

Last time: Given $\{(t_i, f(t_i))\}_{i=1}^{m}$
- $t_1 < t_2 < \dots < t_m$
- f is m times differentiable
- $P_{m-1}(t)$ polynomial interpolant

then $\|f(t) - P_{m-1}(t)\|_\infty = \max_{t \in [t_1, t_m]} |f(t) - P_{m-1}(t)| \le \frac{M h^m}{4m}$

where $M \geq |f^{(m)}(t)|$ for $t \in [t_1, t_m]$ and $h = \max |t_{i+1} - t_i|$

Wed 12 May 2025

A reasonable question to ask is:

"If we increase the number of interpolation points while keeping $(t_m - t_1)$ (interval width) constant, will the approximation error necessarily go down?"

We have error $\le \boxed{\frac{1}{4m}} \cdot M \cdot h^m$

- $\frac{1}{4m}$: decreases as m increases
- M: behaviour depends on $f(t)$
- h^m: when $h < 1$, decreases as m increases

Let's look at two examples.

① $f(t) = \sin(\pi t)$

- interpolate at equally spaced points in $[-1, 1]$.

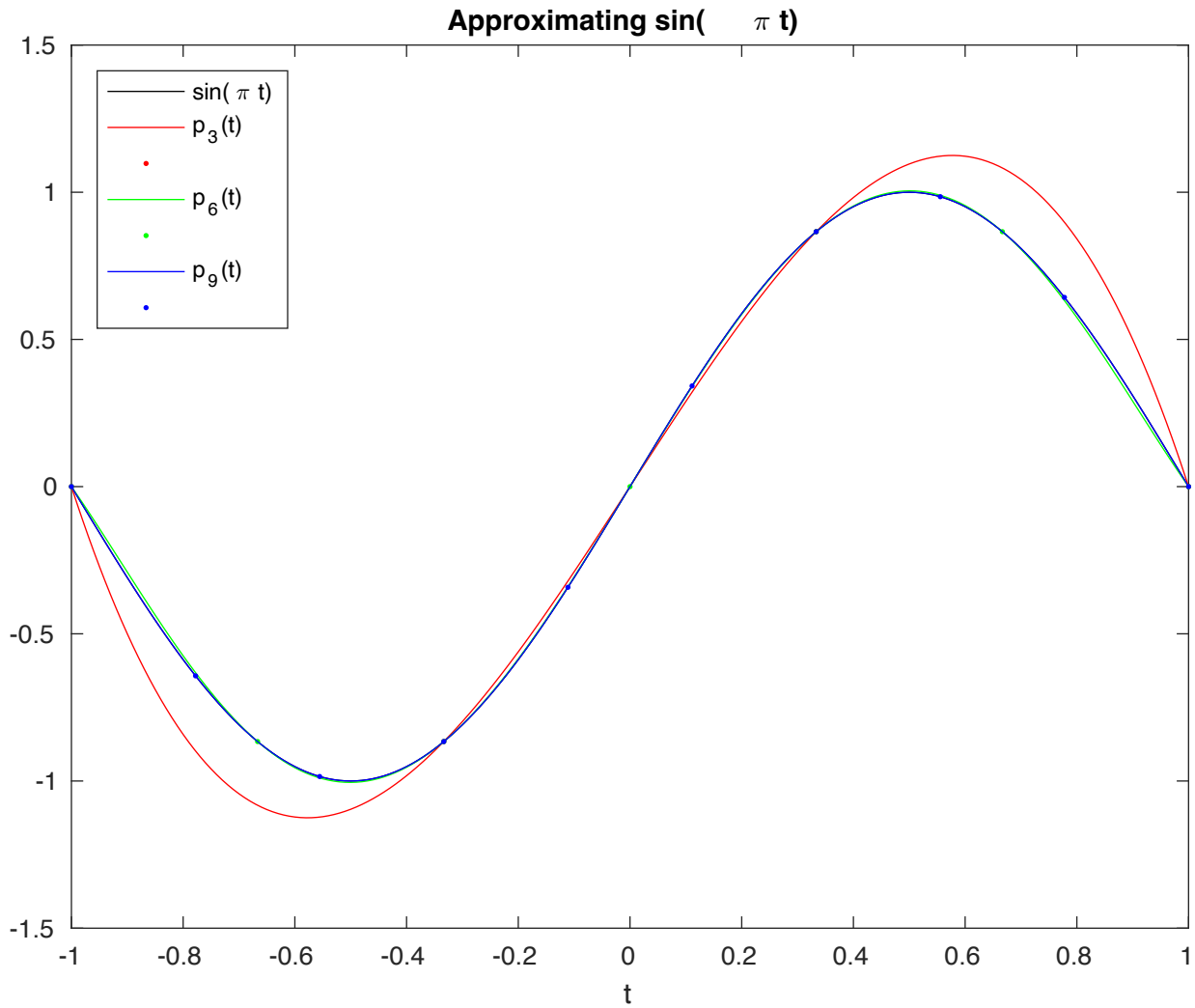
$m = 4$ $\quad t_i \in \{-1, -1/3, 1/3, 1\}$ get $P_3(t)$

$m = 7$ $\quad t_i \in \{-1, -2/3, -1/3, 0, 1/3, 2/3, 1\}$ get $P_6(t)$

$m = 10$ $\quad t_i \in \{-1, -7/9, \dots 7/9, 1\}$ get $P_9(t)$

Note: plots

- observe that the approximation gets more accurate with increasing m


Approximating sin(π t)
sin(π t)
p3(t)
p6(t)
p9(t)
1.5
1
0.5
0
-0.5
-1
-1.5
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
t

<u>note</u>: $\max_{t \in [-1,1]} |f^{(m)}(t)| = \pi^m$ which grows slower than m!

Example 2

Consider $f(t) = \dfrac{1}{1+25t^2}$ (Runge's function)

(looks a bit like Normal distribution)

- interpolate using equally spaced points in $[-1,1]$.

$m=4 \rightarrow P_3(t)$
$m=7 \rightarrow P_6(t)$
$m=10 \rightarrow P_9(t)$
$m=13 \rightarrow P_{12}(t)$

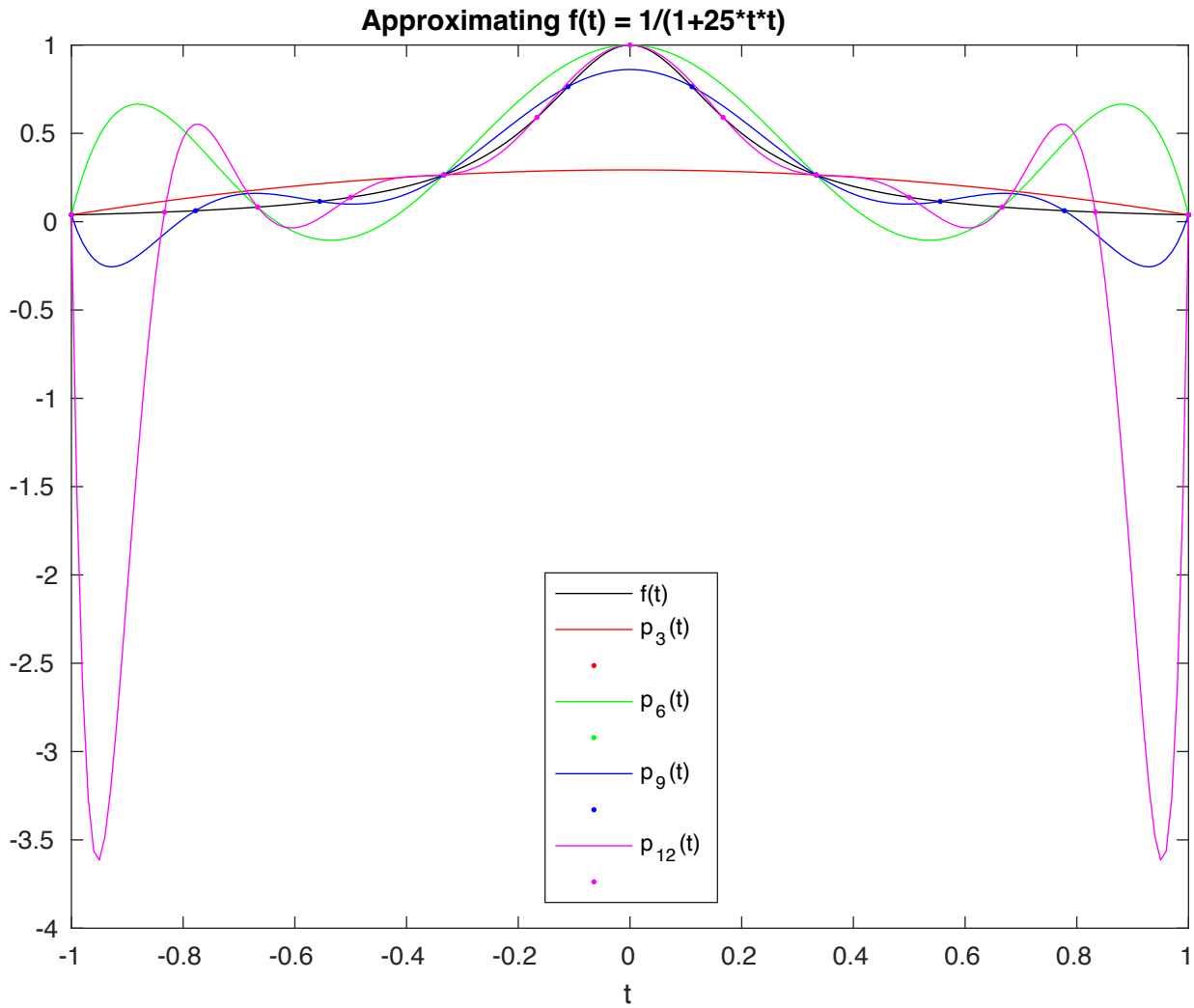
note: plots.

- in the middle area, the approximation does become more accurate as m increases.

<u>but</u> at the end points, the error grows with m.

<u>Lesson</u>: Degree $m-1$ polynomial interpolants do not necessarily converge to the function that they interpolate as m (the # of interpolation points) increases!

Q: So what should you do if you need to interpolate a large data set?!


Approximating f(t) = 1/(1+25*t*t)
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
-4
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
t
f(t)
p3(t)
p6(t)
p9(t)
p12(t)

Piecewise Polynomial Interpolation

Solution idea: Instead of constructing a single high degree polynomial that interpolates all of the data, construct a set of lower degree polynomials ($\hat{d}$)

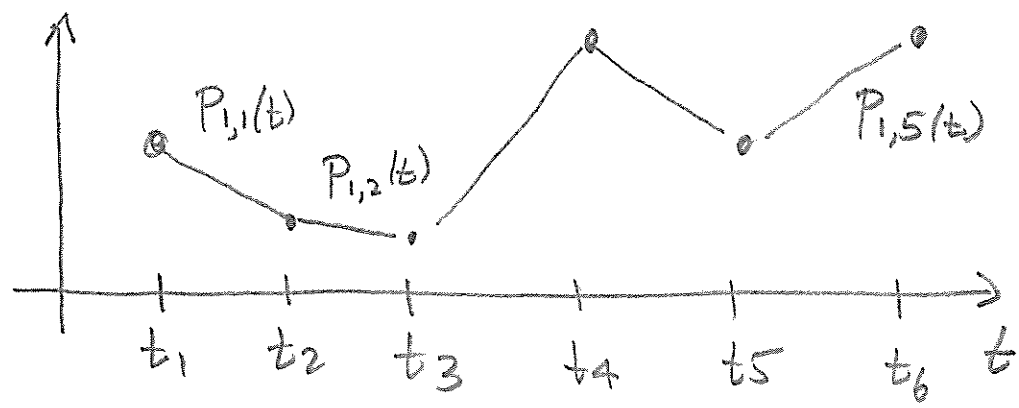
$$\{P_{d,i}(t)\}$$ where each $P_{d,i}(t)$ interpolates a subset of the data.

degree, interval number

e.g. A piecewise linear interpolant $\{P_{1,i}(t)\}_{i=1}^{m-1}$

Given $\{(t_i, y_i)\}_{i=1}^{m}$ with $t_i < t_{i+1}$,

Connect the points with straight lines.



The interpolant is $L(t) = \{P_{1,i}(t)$, when $t \in [t_i, t_{i+1}]$

The $\{t_i\}_{i=1}^{m}$ are called the Knots of the piecewise interpolant.

Using the Lagrange form of the interpolant, we can immediately write

$$P_{1,i}(t) = y_i \frac{(t-t_{i+1})}{(t_i-t_{i+1})} + y_{i+1} \frac{(t-t_i)}{(t_{i+1}-t_i)}$$

(or, via Newton, $P_{1,i}(t) = y_i + \frac{y_{i+1}-y_i}{t_{i+1}-t_i}(t-t_i)$

Note: $L(t)$ is continuous. Since each piece is linear, we need only worry about the connection at the knots.

$$P_{1,i}(t_{i+1}) = y_{i+1}$$
$$= P_{1,i+1}(t_{i+1})$$

but $L'(t)$ is <u>not</u> continuous

$$P_{1,i}'(t) = \frac{y_{i+1}-y_i}{t_{i+1}-t_i} \qquad P_{1,i+1}'(t) = \frac{y_{i+2}-y_{i+1}}{t_{i+2}-t_{i+1}}$$

(note: plot has kinks at knots)

What about accuracy?

Suppose $L(t)$ interpolates $f(t)$ at the knots $\{t_i\}_{i=1}^{m}$. Then, for $t \in [t_i, t_{i+1}]$,

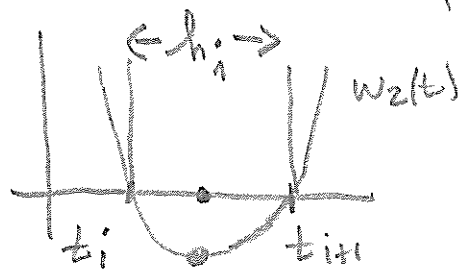
$$f(t) - L(t) = f(t) - P_{1,i}(t)$$
$$= \frac{1}{2!} f''(c_t) \overbrace{(t-t_i)(t-t_{i+1})}^{W_2(t)}$$

for some $c_t \in (t_i, t_{i+1})$.

from before, with $m=2$

$$w_2(t) = (t - t_i)(t - t_{i+1})$$
$$= t^2 - (t_i + t_{i+1})t + t_i t_{i+1}$$

then $w_2'(t) = 2t - (t_i + t_{i+1})$

$= 0$ for $t = \dfrac{t_i + t_{i+1}}{2}$ (midpoint of the interval)

picture: